Mark Scheme Summer 2008

AEA

AEA Mathematics (9801)

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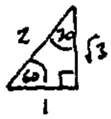
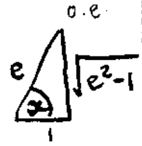
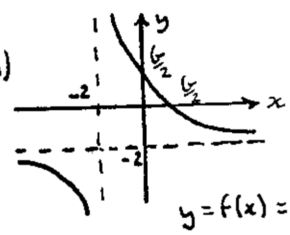
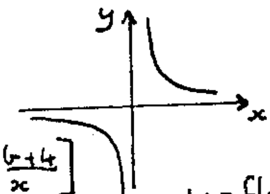
Contents

AEA June 2008	NOTES	MARKS
① $a=200,\ d=-\frac{5}{2}$ $\quad u_n=0 \Rightarrow 200-\frac{5}{2}(n-1)=0$	Identify a, d and set $u_n=0$	M1
$\Rightarrow \quad n=81$		A1
[ALT $S_n=\frac{n}{2}\left(400-\frac{5}{2}(n-1)\right)$  Max at 80.5	S_n and attempt max	M1 A1]
Maximum sum when $n=80$ or 81	Use of S_n with their $n=80$ or their $n=81$	M1 A1
$S_{80}=40\left[400-\frac{5}{2}\times 79\right]$		
$=20\left[800-395\right]$		
$=\underline{\underline{8100}}$		A1 ⑤
② (a) $\frac{dy}{dx}=2 \Rightarrow 2(x+1)(x+2)=xy$	Sub $\frac{dy}{dx}=2$	M1
$\Rightarrow 2(x^2+3x+2)=x(2x+5)$	sub y for $2x+5$ and attempt to solve	M1
$y=2x+5 \Rightarrow$ $x=-4$		A1
$y=-3$ [or P is $(-4,-3)$]		A1 (4)
(b) $\int \frac{1}{y}\,dy = \int \frac{x}{(x+1)(x+2)}\,dx$	Separation attempt	M1
$=\int\left(\frac{2}{x+2}-\frac{1}{x+1}\right)dx$	Attempt partial fractions	M1
$\Rightarrow \ln\lvert y\rvert = 2\ln\lvert x+2\rvert - \ln\lvert x+1\rvert\ (+c)$	Some correct ln integral of x function	M1 A1
$\ln y = \ln\left[\frac{A(x+2)^2}{(x+1)}\right]$ or $\ln\left[\frac{(x+2)^2}{(x+1)}\right]+c$	Use of log rules $\to \ln[g(x)]$ (condone $A=1$ or $c=0$)	M1
$y=\frac{A(x+2)^2}{(x+1)}$	Getting out of logs (must have 'A' or equiv)	M1
Using $P(-4,-3) \Rightarrow -3=\frac{A(-2)^2}{(-3)}$ $\quad$ ($\checkmark$ their P)	Use P to form eqn in A or C	M1
$y=\frac{9(x+2)^2}{4(x+1)}$		A1 (8)
	Max S1 only for 11 or 12/12	⑫

S1 or S2	For a fully correct (or nearly so) and neat or succinct solution to Qu2 - Qu7. Count best 3 questions.
T1	For a good attempt at the whole paper (all questions)

AEA June 2008	Notes	Marks
3 (a) $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$; $t = \tan 15$  	Use of ~~known~~ ...	B1
$\tan 30 = \frac{1}{\sqrt3} \Rightarrow \frac{1}{\sqrt3} = \frac{2t}{1-t^2}$	Use ~~tan 2θ~~ → equation in t	M1
$t^2 + 2\sqrt3 t - 1 = 0 \Rightarrow t = \frac{-2\sqrt3 \pm \sqrt{12+4}}{2}$	Attempt to solve → t=	M1
$t = \tan 15 = 2 - \sqrt3$ (*)	[S for considering ±]	A1 cso (4)
(b) $\left(\frac{\sin\theta}{2} + \frac{\sqrt3}{2}\cos\theta\right)\left(\frac{\sin\theta}{2} - \frac{\sqrt3}{2}\cos\theta\right) = \cos^2\theta(1-\sqrt3)$	Use of sin(A ± B)	M1
$\frac{\sin^2\theta}{4} - \frac34\cos^2\theta = \cos^2\theta - \sqrt3\cos^2\theta$	Equation in s^2 and c^2 or c^2 and cos2θ	M1
$\frac{\sin^2\theta}{4} = \frac{\cos^2\theta}{4}(7-4\sqrt3)$	Attempt cos²θ, ~~tan²θ~~ or cos2θ or sin²θ	M1
		A1
$\cos^2\theta = \frac{1}{4(2-\sqrt3)}$ or $\frac{2+\sqrt3}{4}$ or $\tan^2\theta = 7 - 4\sqrt3$ or $\cos 2\theta = \frac{2\sqrt3-3}{4-2\sqrt3}$	$(2-\sqrt3)^2 = 7-4\sqrt3$	M1
$\tan^2\theta = (2-\sqrt3)^2$		
$\tan\theta = \pm(2-\sqrt3)$ or $\cos 2\theta = \frac{\sqrt3}{2}$		A1
$\tan\theta = 2-\sqrt3 \Rightarrow \theta = 15, 195$; $\tan\theta = -(2-\sqrt3) \Rightarrow \theta = 165, 345$		A1; A1 (8) (12)
4. (a) $\frac{dy}{dx} = -\sin x\ln(\sec x) + \cos x.\tan x$	Use of product rule	M1 A1
$y' = 0 \Rightarrow 0 = \sin x(1 - \ln(\sec x))$	Take out sin x factor	M1
[$\sin x = 0 \Rightarrow x = 0$ ∴ Min at origin]	[S marks]	
$\ln\sec x = 1 \Rightarrow \sec x = e$; ∴ B is $\left(\arccos\frac1e, \frac1e\right)$		A1 ; A1 (5)
(2) [Rectangle − S]	For strategy	M1
(b)  $I = \int \cos x\ln(\sec x)dx = \sin x\ln\sec x - \int \sin x\tan x\,dx$	Attempt parts	M1 A1
$I = \sin x\ln\sec x - \int\frac{\sin^2 x}{\cos x}dx = \sin x\ln\sec x - \int(\sec x - \cos x)dx$	Put sin x tan x into integrable form	M1
$I = \sin x\ln\sec x - \ln\lvert\sec x + \tan x\rvert + \sin x$	correct integration	A1 A1
$S = [I]_0^{\arccos\frac1e}$ 	Attempt correct limits and sin x and cos x in terms of e, $\sqrt{e^2-1}$ etc.	M1 Must have complete integral
$S = \frac{\sqrt{e^2-1}}{e} - \ln\left[e + \sqrt{e^2-1}\right] + \frac{\sqrt{e^2-1}}{e}$		
Area $= 2\left[\frac1e\arccos\frac1e - S\right] = \frac2e\arccos\frac1e + 2\ln\left(e + \sqrt{e^2-1}\right) - \frac{4\sqrt{e^2-1}}{e}$ (*)		A1 cso (8) (13)

	Notes	Marks
5. (i) $(\log_3 p)^2 = \log_3(p^2) \Rightarrow (\log_3 p)^2 = 2\log_3 p$	Use $n\log x$ rule	M1
$\Rightarrow \log_3 p(\log_3 p - 2) = 0 \Rightarrow \log_3 p = 0 \therefore p = 1$		A1
$\log_3 p = 2 \therefore p = 9$		A1
or		
$\log_3(p+q) = \log_3 p + \log_3 q \Rightarrow \log_3(p+q) = \log_3(pq)$	Use of $\log x + \log y$ rule	M1
		A1
$\Rightarrow p+q = pq$ OR $9+q = 9q$	Making q the subject [S for $p \neq 1$]	M1
$\therefore q = \frac{p}{p-1} \quad (p \neq 1)$		A1 (7)
$\underline{p = 9} \Rightarrow \underline{q = \frac{9}{8}}$		
$\log_3 9 = 2$ o.e.	Seen anywhere	B1
(ii) $\log_3\left[\frac{3x^3 - 23x^2 + 40x}{(3x-8)^2}\right] = 1$	Use of log rules to form a single log	M1
	out of logs	M1
$\frac{3x^3 - 23x^2 + 40x}{(3x-8)^2} = 3$	For reducing cubic equation to quadratic [$x \neq 8/3$ for S mark]	M1
$\frac{x(\cancel{3x-8})(x-5)}{(3x-8)^{\cancel{2}}} = 3$; $3x^2 - 44x + 96 = 0$	3TQ (accurate 3TQ)	A1
$x^2 - 5x = 9x - 24 \Rightarrow x^2 - 14x + 24 = 0$		
$(x-12)(x-2) = 0$; $\Rightarrow x = 2$ or $\underline{12}$	(ignore $x = 2$ and $8/3$)	M1 ; A1 (7)
($x = 8/3$ listed here loses final A1)	[S marks for comment on 2, 8/3]	14
6. (a) $yx + 2y = ax + b \Rightarrow x = \frac{b-2y}{y-a} \therefore \underline{f^{-1}(x) = \frac{b-2x}{x-a}}$	Make x the subject	M1, A1 (2)
(b) $ff(x) = x \Rightarrow f^{-1}(x) = f(x)$; $\therefore \underline{a = -2}$	$f^{-1} = f$	M1; A1 (2)
(c) (i)  $y = f(x) = \frac{b-2x}{x+2}$	(i) shape	B1 (no overlap)
	$x = -2$; $y = -2$	B1
	$(\frac{b}{2}, 0)$, $(0, \frac{b}{2})$	B1 (3)
(ii)  $\left[y = \frac{b+4}{x}\right]$ $y = f(x-2) + 2$	(ii) $\rightarrow +2$	M1
	$\uparrow +2$	M1
	both branches	A1 (3)
(d) Normal at P' on $y = f(x-2)+2$ is: $y = 4(x-2) - 39 + 2$	Use transformations on normal	M1
$y = 4x - 45$		A1
$\therefore$ curve is symmetric about $y = \pm x$, normal at Q' will be $y = 4x + 45$ [symmetry is $x \to -x$ and $y \to -y$]	Use symmetry on Q'	M1
Reversing process		
Normal at Q on $y = f(x)$ is: $y = 4(x+2) + 45 - 2$	Use $f(x+2) - 2$	M1
$\therefore \underline{y = 4x + 51 \text{ or } k = 51}$		A1 (5)
		15
[N.B. $P' : (12, 3)$; P is $(10, 1)$ M1A1; $b = 32$; $Q = (-14, -5)$ M1] $y - -5 = 4(x - -14)$	M1	
$\rightarrow k = 51$	A1	

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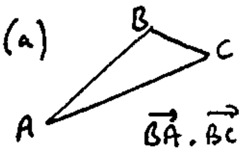
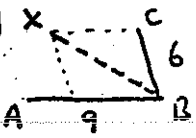
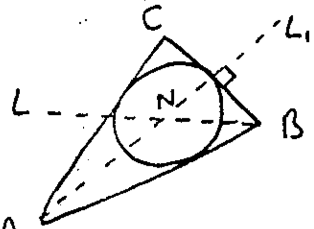
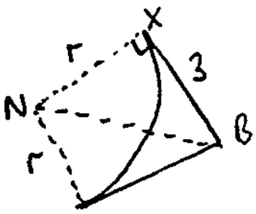
	Notes	Marks
7. (a)  $\overrightarrow{BA} = \begin{pmatrix} -4 \\ 1 \\ -8 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} 4 \\ 2 \\ -4 \end{pmatrix}$	Attempt $\overrightarrow{BA}$ and $\overrightarrow{BC}$	M1
$\overrightarrow{BA}.\overrightarrow{BC} = -16+2+32 = 18$	Attempt $\overrightarrow{BA}.\overrightarrow{BC}$	M1
$\lvert\overrightarrow{BA}\rvert = \sqrt{4^2+1^2+8^2} = 9$, $\lvert\overrightarrow{BC}\rvert = \sqrt{4^2+2^2+4^2} = 6$	Attempt $\lvert\overrightarrow{BA}\rvert$ or $\lvert\overrightarrow{BC}\rvert$	M1
$\therefore \cos\theta = \frac{18}{9\times 6} = \frac{1}{3}$		A1 (4)
(b)  Using Rhombus idea, $\overrightarrow{BX} = \overrightarrow{BC} + \frac{2}{3}\overrightarrow{BA}$ o.e.	eg $3\overrightarrow{BC} + 2\overrightarrow{BA}$ Any correct ratio	M1, A1
$= \lambda\begin{pmatrix} 4 \\ 8 \\ -28 \end{pmatrix}$ or $\mu\begin{pmatrix} 1 \\ 2 \\ -7 \end{pmatrix}$		A1
Through B $\therefore$ $\underline{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + t\begin{pmatrix} 1 \\ 2 \\ -7 \end{pmatrix}$ (*)		A1 cso. (4)
(c) $\overrightarrow{AC} = \begin{pmatrix} 8 \\ 1 \\ 4 \end{pmatrix}$ $\lvert\overrightarrow{AC}\rvert = \sqrt{8^2+1^2+4^2} = 9 = \lvert\overrightarrow{BA}\rvert$	Attempt $\overrightarrow{AC}$ and $\lvert\overrightarrow{AC}\rvert$ (must say $=\lvert\overrightarrow{BA}\rvert$ for A1)	M1 A1 cso (2)
		
(d) $\because$ ABC is Isos L_1 has direction $\frac{1}{2}(\overrightarrow{AB}+\overrightarrow{AC}) = \begin{pmatrix} 6 \\ 0 \\ 6 \end{pmatrix}$		B1
$\therefore L_1$ has equation $(\underline{r} =) \begin{pmatrix} -3 \\ 1 \\ -9 \end{pmatrix} + u\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$	Find equation of L_1	M1
Centre of S is intersection of L_1 and L	Strategy	M1
Solving: $\left.\begin{matrix} 1+t = -3+u \\ 2t = 1 \end{matrix}\right\} \Rightarrow t = \frac{1}{2}$, $u = 9/2$	Attempt to solve $t =$ or $u =$	M1 A1
$\left[\ -1-7t = -9+u \quad \text{Check: LHS} = -\frac{9}{2},\ \text{RHS} = -\frac{9}{2}\ \right]$		
$\therefore$ Centre has position vector $\overrightarrow{ON} = \begin{pmatrix} 3/2 \\ 1 \\ -9/2 \end{pmatrix}$	(-1eoo)	A2/1/0 (7)
(e)  Let X = mid-point of BC $\therefore BX = 3$ ($\because$ Isos)	BX = 3	B1
$\overrightarrow{BN} = \begin{pmatrix} 1/2 \\ 1 \\ -7/2 \end{pmatrix}$ $\therefore \overrightarrow{BN} = \frac{1}{2}\sqrt{54}$	Attempt $\overrightarrow{BN}$ & $\lvert\overrightarrow{BN}\rvert$	M1 A1
$r^2 = BN^2 - 3^2$ $\therefore r^2 = \frac{54}{4} - 9$. $\therefore r = \frac{\sqrt{18}}{2}$ or $\frac{3}{2}\sqrt{2}$	Full method for r	M1 A1 (5)
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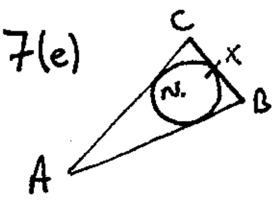
AEA June 2008 ALTERNATIVE Solutions	Notes	Marks
3(a) $\tan 15 = \tan(60-45) = \frac{\tan 60 - \tan 45}{1+\tan 60 \tan 45} = \frac{\sqrt{3}-1}{1+\sqrt{3}}$	or $\tan(45-30)$	B1 M1
$= \frac{(\sqrt{3}-1)^2}{3-1}$, $= \frac{4-2\sqrt{3}}{2} = \underline{2-\sqrt{3}}$	NB no need for $\pm$ consideration this way	M1, A1 (4)
3(b) Use of $\cos A - \cos B = \cdots\cdots = -2\sin\left(\frac{2\theta+120}{2}\right)\sin\left(\frac{2\theta-120}{2}\right)$		M1
$\rightarrow -\frac{1}{2}\left[\cos 2\theta - \cos 120\right] = (1-\sqrt{3})\cos^2\theta$	See snippets 3A	
$\rightarrow \cos 2\theta + \frac{1}{2} = 2(\sqrt{3}-1)\cos^2\theta$		M1
May then go for $\cos^2\theta = \cdots$ or $\cos 2\theta = \cdots$ as per scheme.		
6(d) For a valid method $\rightarrow P=(10,1)$ or $P'(12,3)$		M1 A1
For obtaining $Q=(-14,-5)$	Requires $b=32$ on the way!	M1
Equation of normal: $y - -5 = 4(x - -14)$		M1
$y = 4x+51$ or $k=51$		A1 (5)
7(e)  Midpoint of BC, $\vec{OX} = \begin{pmatrix} 3 \\ 1 \\ -3 \end{pmatrix}$	Must be in (e)	B1
$\vec{NX} = \pm\begin{pmatrix} 3/2 \\ 0 \\ 3/2 \end{pmatrix}$		M1 A1
$r = \lvert\vec{NX}\rvert = \sqrt{\left(\frac{3}{2}\right)^2+\left(\frac{3}{2}\right)^2} = \underline{\underline{\frac{3}{2}\sqrt{2}}}$ o.e.		M1 A1 (5)
7(d) $\vec{ON} = \begin{pmatrix} 1+t \\ 2t \\ -1-7t \end{pmatrix}$	General $\vec{ON}$ =	B1
$\vec{AN} = \begin{pmatrix} 4+t \\ -1+2t \\ 8-7t \end{pmatrix}$	Attempt $\vec{AN}$ in t	M1
$\vec{AN}\cdot\vec{BC} = 0 \Rightarrow 16+4t-2+4t-32+28t=0$	$\vec{AN}\cdot\vec{BC}=0$	M1
$36t = 18$	Solve for t	M1
$t = \frac{1}{2}$	Then as before.	
2(b) $\ln\lvert y\rvert = \frac{1}{2}\int \frac{2x+3}{x^2+3x+2}\,dx - \frac{3}{2}\int \frac{1}{(x+\frac{3}{2})^2-(\frac{1}{2})^2}\,dx$	Sep Equiv to PIF	M1 M1
$= \frac{1}{2}\ln\lvert x^2+3x+2\rvert - 3\ln\left\lvert\frac{x+\frac{3}{2}-\frac{1}{2}}{x+\frac{3}{2}+\frac{1}{2}}\right\rvert \ (+c)$	Correct ln integration	M1 A1
$= \cdots$	etc.	

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